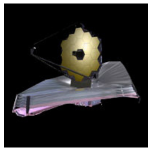

# 10967 - CLEARER: Characterizing Light Extinction And Reddening using Emission-line Ratios

Cycle: 5, Proposal Category: AR

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**OBSERVATIONS**

**ABSTRACT**

Dust attenuation curves are essential for accurately interpreting galaxy properties and modeling galaxy evolution. Variations in these curves introduce large uncertainties, up to a factor of 3 in stellar mass and 10 in star formation rate, and impact photometric redshift accuracy, a key requirement for cosmological surveys with Rubin, Euclid, and Roman. Public data realeases of deep NIRSpec MSA spectroscopy now enable new constraints on dust attenuation using Paschen emission lines, which trace dust reddening even in galaxies with high optical depths that obscure commonly used diagnostics based on the Halpha and Hbeta lines.

We propose CLEARER (Characterizing Light Extinction And Reddening using Emission-line Ratios) to derive dust attenuation curves from Paschen line ratios at z ~1.6, when cosmic star formation is near its peak and most of it is heavily dust-obscured. Using deep NIRSpec/prism spectroscopy and HST UV/optical imaging from the public JWST surveys, we will select galaxy samples with S/N > 5 Paschen line detections. These data will provide rest-frame coverage of the UV slope, the 2175Å absorption feature, and Paschen line ratios, offering a uniquely powerful view of dust in dusty high-redshift galaxies. CLEARER will deliver the first Paschen line-derived dust attenuation curves at z~1.6, informing models of galaxy evolution, and enabling more accurate measurements for upcoming large-scale surveys.

**OBSERVING DESCRIPTION**